

GRADUATION PROJECT — AI ENGINEERING BOOTCAMP

The Student's Channel Brain

Benchmark & LangSmith Analysis Report

MIT 18.06 Linear Algebra Tutor — 5-Lecture Retrieval Benchmark Stage

Benchmark questions	8 retrieval questions (R1–R8)
Configurations compared	Chunk size · retrieval k · reranker OFF/ON
LangSmith project	students-channel-brain
Evidence sources	Notebook 05 execution + exported LangSmith benchmark_question traces

1. Executive Summary

Purpose. The benchmark isolated three retrieval decisions: chunk size, vector-retrieval k, and CrossEncoder reranking. The evaluation metric was whether the top retrieved chunk came from the expected lecture, with latency recorded for each run.

Observed benchmark decision. The notebook selected 1200-character chunks, k=3, and reranker ON. The strongest measurable gain came from chunk size: 1200 reached a 0.625 top-lecture accuracy rate versus 0.500 for 600. k=3 and k=5 tied at 0.625. Reranker OFF and ON also tied at 0.625, while reranking increased average retrieval latency from about 0.0163 s to 0.0444 s.

2. Benchmark Methodology

The notebook used a one-change-at-a-time comparison around a fixed baseline: 1200-character chunks, k=3, reranker OFF. Eight retrieval questions were used, each with an expected lecture label.

Experiment	Configuration A	Configuration B	What was isolated
Chunk size	1200 / k=3 / reranker OFF	600 / k=3 / reranker OFF	Effect of chunk granularity
Retrieval k	1200 / k=3 / reranker OFF	1200 / k=5 / reranker OFF	Effect of retrieving more candidates
Reranker	1200 / k=3 / reranker OFF	1200 / k=3 / reranker ON	Effect of CrossEncoder ordering

Top-lecture correctness = 1 when the top retrieved document metadata matched the expected lecture; otherwise 0. Latency was measured inside the retrieval/reranking function. For reranker runs, reranker latency was also recorded separately.

Memory and out-of-scope questions were present in the master test set but were intentionally not part of the parameter loops. This benchmark therefore evaluates retrieval configuration, not end-to-end memory or scope-guard behavior.

3. Results

Configuration	Questions	Top-lecture accuracy	Avg latency (s)	Avg reranker latency (s)
Chunk 1200, k=3, OFF	8	0.625	0.0120	0.0000
Chunk 600, k=3, OFF	8	0.500	0.0099	0.0000
Chunk 1200, k=3, OFF (k test)	8	0.625	0.0249	0.0000
Chunk 1200, k=5, OFF	8	0.625	0.0172	0.0000
Chunk 1200, k=3, reranker OFF	8	0.625	0.0163	0.0000

Configuration	Questions	Top-lecture accuracy	Avg latency (s)	Avg reranker latency (s)
Chunk 1200, k=3, reranker ON	8	0.625	0.0444	0.0292

3.1 Chunk Size Decision

1200-character chunks outperformed 600-character chunks on top-lecture accuracy (0.625 vs 0.500). The latency difference was small (about 2.2 ms on average in this run). R2 and R4 were concrete examples where 1200 returned the expected lecture while 600 did not. The notebook therefore selected 1200.

3.2 Retrieval k Decision

k=3 and k=5 produced the same 0.625 top-lecture accuracy. The k=5 run happened to be faster in this single small sample, but the notebook correctly did not treat that as strong evidence because retrieval latency at this scale is noisy. Since no quality gain was observed, k=3 was selected at benchmark time for lower downstream context volume.

3.3 Reranker Decision

Reranker OFF and ON both produced 0.625 top-lecture accuracy. Reranking changed ordering and improved some individual cases, but it also hurt at least one case. Average latency increased from 0.0163 s to 0.0444 s, with about 0.0292 s attributable to reranking. The reranker remained ON because it was part of the intended architecture and demonstrated measurable ranking impact.

4. Case-Level Analysis of the Selected Benchmark Configuration

Under the benchmark-time selected configuration (1200 / k=3 / reranker ON), five of eight questions had the expected lecture as the top result. Three cases failed: R2, R5, and R8.

Case	Expected	Top result	Pass?	Interpretation
R1	L1	L1	Yes	Row-picture query retrieved Lecture 1 correctly.
R2	L1	L3	No	Reranking promoted a Lecture 3 chunk above relevant Lecture 1 chunks.
R3	L1	L1	Yes	Column-picture linear-combination query retrieved Lecture 1.
R4	L2	L2	Yes	Elimination query retrieved Lecture 2.
R5	L3	L2	No	Explicit Lecture 3 request was not enforced by metadata filtering.
R6	L4	L4	Yes	Explicit LU question retrieved Lecture 4.
R7	L4	L4	Yes	Reranker improved this case from L2 (OFF) to L4 (ON).
R8	L1	L3	No	Comparison query retrieved only Lecture 3 chunks in the top 3.

5. LangSmith Trace Analysis

The export contains 64 benchmark_question traces. There are 48 unique case/configuration combinations (6 configurations × 8 questions). The extra 16 traces come from chunk_1200 and chunk_600 being run twice; for the summary above, the most recent run per case/configuration was used.

Each benchmark trace exposes the full question and configuration inputs, plus expected lecture, top lecture, correctness, total latency, reranker latency, retrieved lecture sequence, timestamps, and evidence snippets. This is sufficient to explain not only whether a case failed, but how retrieval ordering led to the failure.

The exported traces contain no recorded benchmark errors. The LangSmith setup therefore provided stable observability for the benchmark execution itself.

The most visible systematic latency effect was reranking: approximately 29 ms average reranker time, increasing total retrieval-stage latency by roughly 28 ms compared with reranker OFF in this small run.

6. Failure Analysis and Limitations

Failure pattern	Evidence	Likely cause	Future mitigation
Explicit lecture request not enforced	R5 asks for Lecture 3 but top result was Lecture 2	Semantic similarity alone does not guarantee lecture-number compliance	Apply lecture_number metadata filtering when the user explicitly names a lecture.
Multi-concept comparison retrieval	R8 expected Lecture 1 but top 3 were Lecture 3	Single semantic query may over-focus on one related concept/chunk family	Use query decomposition or gather evidence separately for both concepts.
Reranker can help or hurt individual cases	R7 improved; R2 worsened	CrossEncoder reorders only the candidate set it receives	Increase candidate pool before reranking and validate on broader final test set.
Small evaluation set	Only 8 retrieval questions	Metrics are sensitive to a few cases and do not represent all 34 lectures	Use the 38-task end-to-end evaluation for final system-level conclusions.

The reranker experiment used $k=3$ and $\text{rerank_top_n}=3$. This tests ranking impact but gives the reranker no opportunity to select a better document that was outside the initial top 3. Therefore, the benchmark should not be interpreted as a strong test of reranker recall improvement.

7. Conclusions

- 1200-character chunks were the clearest benchmark winner and remain the final chunking choice.
- The original $k=3$ vs $k=5$ comparison showed no top-lecture accuracy difference in the 8-question benchmark.
- Reranking changed evidence ordering and added measurable latency, but did not improve aggregate top-lecture accuracy in the original $k=3/\text{top-3}$ setup.

- LangSmith traces made configuration inputs, retrieved lectures, timing, timestamps, and evidence inspectable at question level, enabling concrete failure analysis.
- The main remaining retrieval weaknesses were explicit lecture targeting and multi-concept comparison queries.
- Final production retrieval is retrieve 8 → rerank → keep 3, followed by the 38-task end-to-end validation across all 34 lectures.

Appendix A. Evidence Used

This report was produced from:

- 05_Configuration_Benchmark_LangSmith.ipynb — benchmark code, fixed questions, decisions, and recorded rationale.
- langsmith_benchmark_question_runs.csv — 64 exported benchmark_question traces with inputs, outputs, latencies, and errors.